"VISUAL PERCEPTION IN THE WHITE RAT"

WILLARD L. VALENTINE

Psychological Laboratory, Ohio Wesleyan University

Higginson has recently published an article under this same caption in which he concludes that "rats . . . . employ visual apprehension in a novel manner" (1). He reviews the literature on the use of visual cues by the white rat and finds that there is evidence for the generalization that the "readjustment of an animal to environmental changes may be characterized by a suddenness not explainable in terms of 'frequency' and 'recency.'" His experimental work on the rat is cited to support this contention.

Briefly, his experimental procedure as reported was as follows: A standard circular maze was modified by placing a sliding door between alleys 2 and 3. In running alley 2 the rat was forced into a cul-de-sac. Upon returning it was possible to turn to the left into alley 3, for the door had been opened in the meantime. The rats were given 100 runs in this maze, being forced each time to run the full length of alley 2 before passing into alley 3—and to food. Six feet was thus added to the minimum distance from the starting box to the food box.

After 100 runs the animals were again placed in the maze, but this time the door into alley 3 was left open from the beginning. It might be supposed that the rats would run past the door regardless of the fact that it was now open and complete the run as formerly by running into the cul-de-sac formed by alley 2. If, however, the rats are capable of a "sudden readjustment" we would expect that they might enter alley 3 directly thereby suddenly eliminating 6 feet from the length of the maze. Higginson concludes in favor of the latter statement upon the basis of the following data: Out of a total of 9 rats, 5 went through the open door, completed the remainder of the maze without error, thereby

suddenly eliminating 6 feet of the path. The other 4 rats went past the open door the first time—but "immediately shifted."

This experiment is essentially a repetition of Higginson's work. We have used 19 rats instead of 9, given them a longer training period, and continued the experiment beyond one trial with the door open in order to better understand the phrase "immediately shifted."

A maze was constructed as illustrated. The arrows point out the direction the rat followed. A sliding door at *E* remained closed as the rat ran past on the way to *C*. The alley at *C* led

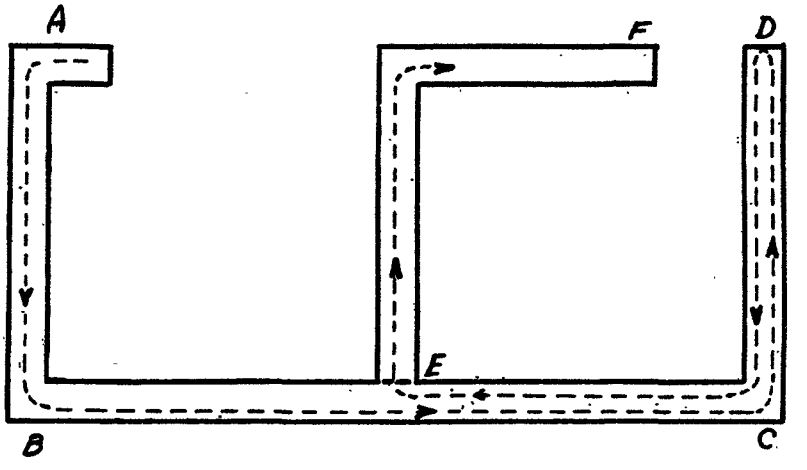


FIG. 1

into a cul-de-sac at *D* and as the rat returned, the door at *E* was raised permitting the animal to obtain food at *F*. The door was operated by a cord and the experimenter was out of the rat's field.

THE TRAINING PERIOD

The rats were given ten runs a day for 24 to 35 days. The total number of runs for each rat are given in Table 1. From the starting box to the end of the cul-de-sac and back to the door constituted a run. If the animal had not run entirely to the end of the cul-de-sac the door was not opened, the run was recorded as incomplete and the rat was placed again in the starting box. If the

animal had come within six inches of the end of the cul-de-sac the door was opened *after he had turned and started back.* He then turned to his right through the open door and ran the remainder of the maze to the food box. This run was recorded as complete. Table 1 shows the number of complete runs in each 10 that composed the day's task.

We had hoped to continue the training period to the point where the rats would make 100 per cent complete runs on a number of consecutive days. This was finally deemed impossible due to the fact that the rats would not run the entire length of the cul-de-sac. Early in the training period they would turn at *C* and return to the closed door. Later they were forced half way up the cul-de-sac. Finally they approached almost to the end. Six inches from the end we had placed an arbitrary mark, beyond which the rat must go in order to find the path to the food clear upon his return. The low percentage of complete runs near the end of the experiment is due to the fact that the rats failed in this respect by an amount which is to be measured in inches. The gross shortenings of the path, which occur when the rat turned at *C* for instance, were no longer apparent. Higginson does not state his experience in this connection, although we should have foreseen this difficulty because of a similar observation in some unpublished work of Dr. F. C. Dockeray. In his experiment a dog was forced to lie down in the process of getting out of a puzzle box. The act of "lying down" deteriorated into a mere ghost of the complete act. Perhaps from Higginson's point of view this is a trivial matter, as the virtue of the cul-de-sac is to get the rat past the door—how far is of little importance, but with our method of recording data it is a matter of major importance.

The 19 rats were of three litters, I, II, III, which were respectively 64, 56, and 41 days old at the beginning of the experiment. They were divided into two groups, as a matter of convenience. The first group was run at five in the afternoon and the second between nine and ten at night. Except for the difference in time the conditions were exactly the same for both groups. A 200-watt lamp with a reflector was placed at a sufficient height to give practically uniform illumination and was used in both afternoon and evening training periods.

TABLE 1

The number of complete runs in each ten

RAT	DAYS																																	TOTAL RUNS
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	
Group I																																				
I 1	7	3	2	9	7	9	9	7	6	6	6	9	6	4	4	5	5	6	3	4	7	8	8	4	10	3	8	8	6	8	6	6	6	5	6	350
I 3		5	6	2	2	4	2	4	2	2	2	3	3	2	4	5	5	6	9	5	9	6	4	6	8	5	3	6	8	6	5	9	8	6	8	340
II 2	4	6	3	7	4	3	5	5	5	4	5	5	8	9	8	5	8	9	7	8	9	9	9	8	6	6	9	5	8	7	3	5	9	4	8	350
II 3		6	6	4	4	5	6	8	8	7	7	5	3	5	6	6	8	7	8	5	7	6	6	8	6	8	2	4	7	7	5	6	6	3	8	340
II 8				2	8	3	5	5	8	9	9	3	9	7	8	7	8	5	10	6	7	9	7	4	9	5	7	7	7	7	4	1	5	4	5	320
II 9								3	3	3	3	1	1	4	3	7	5	7	9	10	9	9	8	5	8	8	5	9	8	6	9	7	8	6	9	280
III 2			4	2	3	3	3	4	4	3	4	6	4	6	8	7	9	6	7	8	10	10	8	9	10	9	8	8	7	7	6	7	9	3	5	330
III 5		5	4	3	4	4	3	6	6	8	9	6	5	7	6	5	8	9	4	9	7	6	8	6	10	6	6	6	9	4	5	2	7	6	10	340
Group II																																				
I 4												8	5	5	5	7	5	4	4	4	7	8	8	8	9	5	7	6	6	5	5	6	6	4	4	240
I 5									4	2	4	3	3	5	2	5	5	5	6	5	4	5	8	6	8	9	7	8	7	7	8	9	7	8	5	270
II 1							0	5	3	1	6	9	6	10	10	7	10	9	10	6	3	9	10	8	6	7	7	8	6	10	7	6	8	10	7	290
II 4		6	3	2	5	3	5	2	4	5	3	4	5	3	4	6	9	7	5	9	9	7	9	9	9	9	6	6	10	5	7	8	10	10	8	340
II 5								4	6	4	4	5	1	2	3	4	5	5	5	9	9	10	8	10	10	5	7	5	6	6	8	4	7	5	8	280
II 6									2	2	4	8	8	8	6	10	7	4	4	3	5	3	7	6	4	8	9	8	5	5	8	6	7	4	1	270
II 11							4	5	5	9	9	5	5	4	4	5	6	5	5	6	7	8	7	7	5	8	8	8	9	9	6	7	9	9	5	280
II 13							3	3	5	3	6	3	3	3	5	4	6	5	8	7	9	9	9	9	10	10	8	7	6	7	4	8	10	10	9	280
II 14													4	5	5	6	6	8	5	4	4	9	8	9	9	10	9	7	6	9	3	8	10	10	4	240
III 3				4	4	7	8	3	4	5	7	7	6	7	9	6	6	8	8	9	9	7	8	8	4	8	9	7	8	7	8	6	4	8	10	330
III 7								3	5	2	2	4	3	9	10	7	3	6	5	7	6	9	9	8	10	7	10	10	7	9	8	4	4	4	7	280

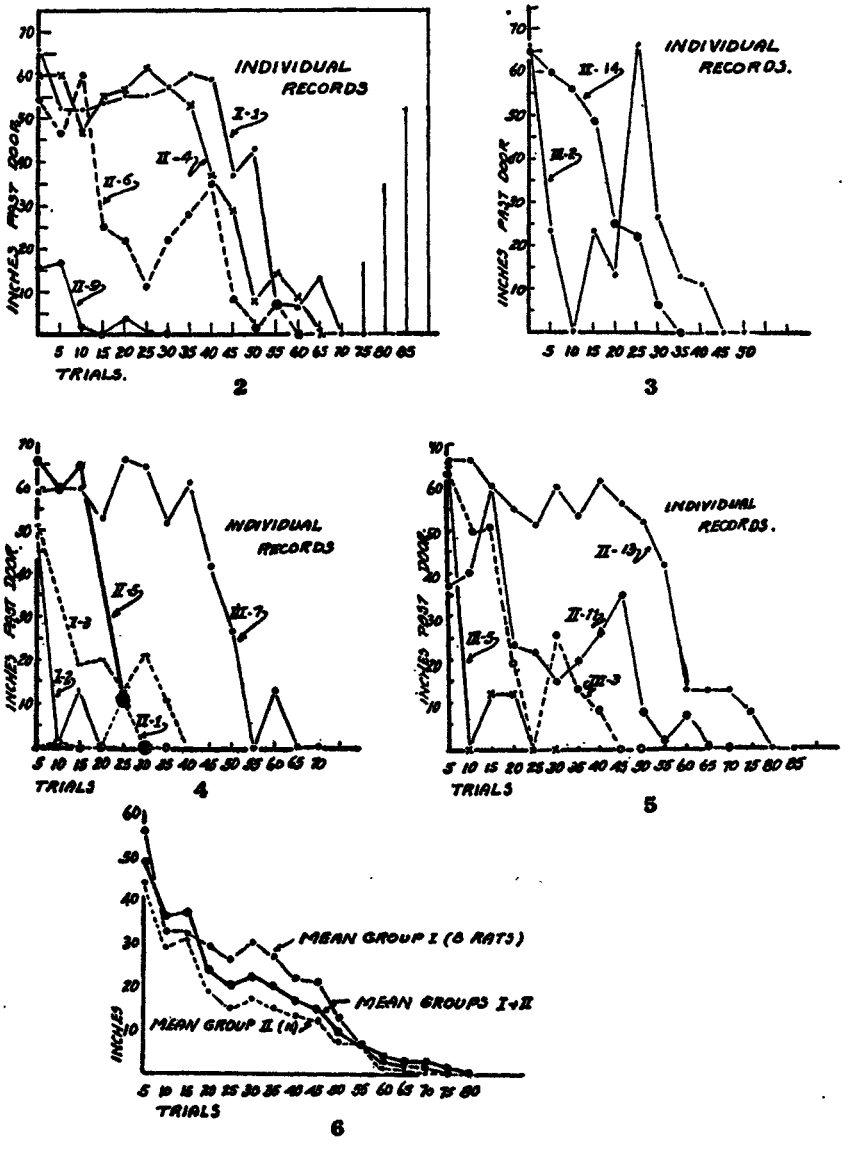
THE CRITICAL RUN

After the completion of the training period, the rats were again placed in the maze, this time with the sliding door open. We were desirous of obtaining moving pictures of this run and because these pictures could not be obtained under the illumination used during the training period, we were forced to make this critical experiment in the afternoon at one—that is, when the sun was brightest and in the proper position to give a maximum illumination in the maze room.

Fearing that this change would in some degree disturb the animals, two normal training runs were given each animal before the critical run. A later examination of the moving picture of these runs showed that no difference could be observed between the behavior of the rats at this time and at the previous regular training periods.

Of the 19 rats 15 ran past the door and 4 went through it when it was opened for the first time (critical run). In carrying Higginson's experiment to its natural conclusion we again placed the rats in the maze for a second critical run. This time 17 went past and 2 entered; two which had entered before ran past this time. A third run gave the same result as the first. In the fourth run 15 ran past and 4 entered. This same procedure was repeated, giving each rat 10 trials a day until at the end of 80-odd trials all of the rats were consistently turning to the left through the open door and had eliminated 132 inches of the path to food.

Quantitative results were secured on the same animals by observing how far past the door they actually ran before turning around. The maze had been marked off in 6-inch intervals and another observer recorded these data. Thus when a rat ran the entire length of the cul-de-sac to *D* (fig. 1) it contributed 66 inches. In returning to *C* it contributed 30 inches. Our assumption in this case is that the distance that the rat runs past the door is somewhat of a measure of its readjustment to the open door. The composite result of the 19 rats is plotted in figure 6 where the ordinates represent the average distance in inches that the rat ran past the door and the abscissa represents the successive runs. All of the points in the graphs are means of five runs.


INDIVIDUAL RECORDS
INCHES PAST DOOR
TRIALS.
2
INDIVIDUAL RECORDS.
3
INDIVIDUAL RECORDS
4
INDIVIDUAL RECORDS.
5
MEAN GROUP I (6 RATS)
MEAN GROUPS I+II
MEAN GROUP II (11)
INCHES
TRIALS
6

FIGS. 2–6

There seems to be some controversy over the interpretation of data of this kind. It is urged that a mean is not representative of the behavior of a group but that the "intelligent man takes the exceptions into account." On the other hand, it appears to the writer that the mean, together with a statistical index representing the deviations of the individual results from this mean is a fair and the only scientific description that can be made, in that it represents the most probable result that will be obtained if the experiment is repeated again under the same conditions. An examination of these individual data indicate that in four instances "sudden learning" was obtained. But rather than to attribute this type of behavior to "apprehensive functions employed in a novel manner" we attribute it to an extraneous factor over which we had no control and which necessitates the employment of a large number of animals in this, or any other experiment. As a matter of fact it was observed that the speed factor operated most in determining whether or not the rat would enter the open door. A fast runner ran past it, but a slow runner was more likely to enter. This observation could very easily be checked by introducing a control in the form of rats which had been fed immediately before the critical run. The outcome of an experiment of this type is fairly obvious.

In conclusion, it is hoped that the result of this experiment is not to be interpretated that rats do not employ the visual function under some very special conditions (2). There seems to be evidence, however, that they do not rely upon vision in any great degree after a habit is once established. As a matter of fact Small (3) reports a case of a blind rat which exhibited this sudden learning and which could hardly be attributed "an extension of visual apprehension."

REFERENCES

(1) HIGGINSON, GLEN: Visual perception in the white rat. Jour. Exper. Psychol., 1926, ix, 337-347.

(2) FIELDS, PAUL E.: Form perception in the white rat. Jour. Comp. Psychol., 1928, viii, 143-158.

(3) SMALL, WILLARD: The mental development of the rat. Amer. Jour. Psychol., 1900, xii, 325.